Project Report: Customer Segmentation Using K-Means

1. Introduction

Customer segmentation is a key strategy for businesses to understand their customers and design targeted marketing campaigns. In this project, we apply **K-means clustering** to the Mall Customers dataset to identify distinct customer groups based on **Annual Income** and **Spending Score**.

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2. Dataset Description

- **File:** Mall_Customers.csv
- **Records:** 200 customers
- **Features:**
  - CustomerID – Unique identifier
  - Gender – Male/Female
  - Age – Age of customer
  - Annual Income (k$) – Annual income in thousands of dollars
  - Spending Score (1–100) – Score assigned based on customer behavior and spending patterns

This dataset is widely used for clustering practice and demonstrates how demographic and behavioral data can be leveraged for segmentation.

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3. Methodology

1. **Data Exploration**
   - Checked distributions of income and spending score.
   - Verified no missing values.

2. **Feature Selection**
    - Focused on Annual Income and Spending Score as clustering variables.
3. **Clustering**
    - Applied **K-means clustering** with $k = 4$.
    - Used the **Elbow Method** to determine the optimal number of clusters.
    - Evaluated cluster quality with **silhouette score**.
4. **Visualization**
    - Scatter plots of clusters with centroids highlighted.
    - Clear separation of customer groups observed.

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5. Results

Age vs Spending Score (4 Clusters)

- Cluster 1 (Red):
  Older customers with moderate spending scores. They represent a segment that spends occasionally but not heavily — potential to nurture into loyal buyers.
- Cluster 2 (Blue):
  Younger to middle-aged customers with high spending scores. This is the most valuable group, showing strong purchasing behavior. They're ideal for premium offerings, loyalty programs, and personalized experiences.
- Cluster 3 (Green):
  Older customers with low spending scores. They appear conservative in spending, possibly less responsive to marketing campaigns. Best approached with cost-efficient engagement strategies.
- Cluster 4 (Cyan):
  Younger to older customers with low spending scores. Likely price-sensitive or early-stage customers. Promotions, discounts, or entry-level products could help convert them into higher spenders.

Annual Income vs Spending Score (5 Clusters)

- Cluster 1 (Red):
  Customers with moderate income and moderate spending scores. They are budget-conscious and unlikely to respond to premium offers. Best suited for affordable products and promotions.
- Cluster 2 (Blue):
  High-income customers with high spending scores.They are premium and high value customers.They should be prioritized with VIP treatment.
- Cluster 3 (Green):
  Low-income customers with high spending scores. They are highly engaged despite not being the wealthiest. Ideal for loyalty programs.
- Cluster 4 (Magenta):
  Low income customers with low spending scores. They are budget consious and value driven.
- Cluster 5 (Cyan):
  High income customers with low spending scores.They have stronf finacial capacity but not engaged much.They could be highly profitable if activated.

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5. Business Insights

- High-Value Segments: Cluster 2 (Age vs Spending) and Cluster 2(Income vs Spending) should be prioritized with exclusive benefits and personalized experiences.
- Growth Potential: Younger, low-spending clusters can be nurtured with promotions, entry-level products, and engagement campaigns.
- Efficiency: Older, low-spending clusters should be approached with cost-effective retention strategies.
- Untapped Potential: High-income but low-spending customers require trust-building, luxury positioning, and personalized engagement.

6.Recommendations

1. Product Strategy: Develop affordable bundles for low-income clusters and premium offerings for high-value clusters.
2. Marketing Strategy: Personalize campaigns by cluster — loyalty rewards for engaged spenders, promotions for price-sensitive groups.
3. Customer Relationship Management: Build trust and exclusivity for high-income low spenders.
4. Retention: Maintain engagement with moderate, steady buyers through consistent value delivery.

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7. Conclusion

K-means clustering successfully grouped mall customers into meaningful segments. This demonstrates the power of unsupervised learning in customer analytics. Future work could include:

- Adding more features like Gender into clustering.
- Trying alternative algorithms (Hierarchical, DBSCAN).
- Evaluating cluster stability across multiple runs